

Aditya Sharma

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EDUCATION

VIT University

Bachelor of Technology in Computer Science and Engineering

DPS Ranipur Haridwar BHEL Uttarakhand

Class XII & Class X

Chennai, Tamil Nadu

Sep. 2022 – May 2026

Haridwar, Uttarakhand

Graduated May 2022

EXPERIENCE

Machine Learning & Computer Vision Developer

2024 – Present

Personal Projects and Research

Remote

- Developed and deployed multiple Computer Vision and Machine Learning applications using PyTorch, YOLOv8/v11, OpenCV, and Flask.
- Built and optimized deep learning pipelines for object detection, video analysis, and edge-device deployment on Raspberry Pi.
- Designed full-stack applications integrating AI models with REST APIs, databases, and real-time video streaming systems.
- Worked extensively with Git and remote-first development workflows while managing end-to-end software projects.

Android Application Developer

2025 – Present

Independent Development

Remote

- Developed PrivacyLens, an offline Android privacy analyzer that inspects installed applications and explains permission risks in plain language.
- Implemented local risk scoring, permission categorization, trusted-app workflows, and direct Android settings integration.
- Designed the application with complete offline processing to ensure user privacy and zero data collection.

PROJECTS

PrivacyLens - Android App Privacy Monitor | Kotlin, Jetpack Compose, DataStore, Room

- Built an offline Android privacy scanner that lists installed applications and separates granted versus requested-only permissions.
- Implemented a local risk engine with score breakdowns, category detection, and trusted-app ignore flow.
- Provided direct Android settings integration to help users review and disable risky permissions safely.

Multi-City GIS-Based ITS Simulator | Python, OSMnx, Gymnasium, Stable-Baselines3, DQN, MARL

- Developed a GIS-grounded autonomous traffic control simulator using real OpenStreetMap road networks.
- Compared No-RL, DQN-based RL, and shared-policy MARL controllers with RSU-assisted V2I communication.
- Evaluated mobility, safety, and infrastructure metrics under communication latency and dropout conditions.

Real-Time Vehicle Accident Detection System | PyTorch, YOLOv11, OpenCV, Python

- Developed a real-time accident detection system using YOLOv11 trained on a custom traffic video dataset.
- Designed a video inference pipeline supporting both live and recorded footage.
- Integrated automated alert triggering for emergency response applications and smart city infrastructure.

Real-Time Airplane Detection & Edge Deployment | YOLOv8, Flask, Raspberry Pi, MySQL

- Trained YOLOv8n on a custom airplane dataset achieving mAP50 of 0.80 and precision of 0.85.
- Deployed the model on Raspberry Pi using Flask-based real-time MJPEG video streaming.
- Integrated MySQL detection logging and automated Excel report generation using openpyxl.

ASL-to-Speech Translator | MediaPipe, OpenCV, Scikit-learn, Amazon Polly

- Built a real-time American Sign Language recognition system using MediaPipe hand landmark extraction.
- Trained a Random Forest classifier achieving 98% accuracy on normalized landmark features.
- Converted recognized gestures into speech using Amazon Polly with an integrated chatbot module.

AI-Generated Video Detection Extension | PyTorch, Flask, CLIP, Chrome Extension

- Engineered a full-stack system to detect AI-generated videos using a Chrome extension and Flask backend.
- Implemented frame sampling with CLIP embeddings and a custom MLP classifier.
- Displayed real-time confidence scores through a browser overlay and supported CPU/GPU auto-detection.

PATENT

Real-Time Vehicle Accident Detection System | *Patent Pending*

- Filed a patent (Application No. 202541054843) for an AI-powered emergency response system using YOLOv11 for real-time accident detection from traffic surveillance footage.

ACHIEVEMENTS

- Solved 250+ Data Structures and Algorithms problems on LeetCode and earned multiple monthly coding badges.
- Qualified for the Smart India Hackathon (SIH) internal selection round.
- Built and deployed multiple end-to-end Machine Learning and Computer Vision projects involving YOLO and Edge AI deployment.

TECHNICAL SKILLS

Languages: Java, Python, Kotlin, C/C++, SQL, JavaScript, HTML/CSS, R

Frameworks: React, Node.js, Flask, FastAPI, Jetpack Compose, REST APIs, Next.js

Machine Learning: PyTorch, YOLOv8, YOLOv11, OpenCV, MediaPipe, Scikit-learn, CLIP, Stable-Baselines3

Developer Tools: Git, Docker, Linux, Google Cloud Platform, Raspberry Pi, Jupyter Notebook, VS Code, PyCharm, IntelliJ

Databases: MySQL, SQLite, Room, DataStore

Libraries: pandas, NumPy, Matplotlib, OSMnx, Gymnasium

CERTIFICATIONS

- Generative AI Using IBM Watsonx – IBM Developer Skills Network
- Databases and SQL for Data Science with Python – Coursera / IBM
- Ethical Hacking Essentials (EHE) – EC-Council